

PROJECT TOPIC: STUDY OF LOCAL FLORA AND FAUNA

Topic Description: Systematic Taxonomic and Ecological Documentation of Ten Prominent Species of Flora and Fauna in Local Surroundings

Biodiversity is not confined to distant Amazonian rainforests or remote national parks; vibrant, complex natural ecosystems exist right in our immediate surroundings—along neighborhood roadsides, institutional campuses, public gardens, rocky scrub hills, and urban riverbanks. Developing keen observational skills to identify, classify, and document local organisms is the foundational starting point for any serious student of environmental science.

This project documents a systematic taxonomic and ecological field survey of **Ten Prominent Species of Flora and Fauna** (5 Plants and 5 Animals/Birds/Insects) inhabiting our local semi-urban landscape in Maharashtra. The survey investigates how indigenous, exotic, and synanthropic species co-exist amidst intense human activity.

By compiling scientific botanical and zoological nomenclature, local vernacular Marathi names, morphological features, ecological niches, and conservation statuses, the project illustrates the vital ecological interdependence between urban vegetation, insect pollinators, avian seed dispersers, and human environmental wellbeing.

2. Importance of the topic (विषयाचे महत्त्व)

Systematic documentation of local biodiversity provides indispensable ecological and conservation insights for the local community:

Why Local Biodiversity Documentation Matters :-

- **Preserving Urban Micro-Ecosystems:** Local biodiversity inventories provide essential baseline data for municipal tree authorities and urban town planners, preventing the mindless destruction of keystone native trees during road expansions.
- **Understanding Mutualistic Trophic Networks:** Highlights how native tree species (such as Peepal and Jamun) sustain essential insect pollinators and fruit-eating birds that cannot survive in artificial monocultures of exotic ornamental trees.
- **Fostering Environmental Empathy & Biophilia:** Stepping outside the classroom to physically observe and sketch living organisms transforms abstract ecological concepts into deep, personal respect for the natural world.
- **Monitoring Climate & Phenological Shifts:** Tracking common neighborhood species over successive years reveals early warning signs of habitat degradation, pesticide overuse, and regional temperature increases.

The documentation proves that even small urban green pockets sustain remarkable ecological diversity.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The field documentation was executed with structured scientific, taxonomic, and ecological objectives:

Structured Project Objectives :-

- **Field Identification & Taxonomy:** To locate, observe, and correctly identify 5 floral and 5 faunal species using standardized botanical and zoological classification keys.
- **Morphological & Habitat Documentation:** To record physical morphological adaptations, life-forms, canopy heights, and specific micro-habitats where each species thrives.
- **Ecological Niche & Mutualism Mapping:** To identify the functional ecological roles of each species—such as primary producer, pollinator, seed disperser, or top carnivore.
- **Anthropogenic Threats Evaluation:** To identify human-induced threats—including concrete paving, pesticide spraying, pruning, and light pollution—affecting each organism.
- **Action Plan for Campus Biodiversity:** To formulate practical guidelines for junior college students to protect and enhance campus wildlife habitats.

The scope encompasses direct morning field transects, visual sketches, and photographic logs.

4. Project Work Methodology (प्रकल्प कार्यपद्धती)

A standardized observational field survey protocol was deployed to record species across a 1-kilometer radius:

Methodological Survey Steps :-

- **Phase 1 – Field Survey Site Selection:** Selected an accessible neighborhood transect encompassing college grounds, an adjacent avenue road, and a municipal botanical garden.
- **Phase 2 – Floral Survey & Botanical Recording:** Examined leaf morphology, bark patterns, floral structures, and fruiting bodies of prominent trees; verified scientific names using standard flora guides.
- **Phase 3 – Faunal Observation & Time Windows:** Conducted early morning (6:30 AM to 8:30 AM) and dusk (5:30 PM to 6:30 PM) observation sessions with 8x40 binoculars to record birds, mammals, and insects.
- **Phase 4 – Taxonomic Cross-Verification:** Cross-verified all scientific classifications, families, and conservation statuses using the IUCN Red List database and Balbharati EVS references.
- **Phase 5 – Report Synthesis:** Compiled findings into standardized tabular records and drafted actionable recommendations.

All organisms were observed in their natural habitats without disturbing or capturing specimens.

5. Observations (निरीक्षणे)

Field transects recorded ten prominent species of flora and fauna, detailed with their taxonomic and ecological characteristics below:

5.1 Tabular Record of 10 Local Flora & Fauna Species

Local & Common Name	Scientific Name & Family	Key Morphological Features	Ecological Niche & Importance
Peepal (पिंपळ)	<i>Ficus religiosa</i> (Moraceae)	Large deciduous tree, heart-shaped leaves with distinct extended drip-tips.	Keystone species; produces fig syconia feeding parakeets, barbets, and fruit bats round the year.
Neem (कडुनिंब)	<i>Azadirachta indica</i> (Meliaceae)	Medium-sized evergreen, pinnate serrated leaves, fragrant white flowers.	Natural bio-pesticide, deep soil aerator; emits antimicrobial phytoncides into local urban air.
Karanj (करंज)	<i>Pongamia pinnata</i> (Fabaceae)	Medium tree, glossy dark green leaflets, lilac-pink pea-shaped blossoms.	Nitrogen-fixing leguminous tree; larval host plant for the Common Banded Awl butterfly.
Adulsa (अडुळसा)	<i>Justicia adhatoda</i> (Acanthaceae)	Perennial dense shrub, lanceolate leaves, tubular white hooded flowers.	Rich in vasicine alkaloid; valuable herbal medicine for respiratory ailments; dense pollinator cover.
Gulmohar (गुलमोहर)	<i>Delonix regia</i> (Fabaceae) [Exotic]	Spreading umbrella canopy, bipinnate leaves, brilliant scarlet-orange blossoms.	Exotic ornamental shade tree; provides urban summer visual beauty, though roots are shallow and storm-brittle.
Red-vented	<i>Pycnonotus cafer</i>	Brownish-black bird, slight crest, scale	Essential frugivore and seed disperser for wild

6. Analysis of data (माहितीचे विश्लेषण)

Ecological analysis of the survey data reveals critical patterns of urban wildlife survival and anthropogenic vulnerability:

Core Analytical Inferences :-

- **Keystone Ecological Role of Ficus Trees:** The survey highlights that indigenous Ficus species (like *Ficus religiosa*) are absolute ecological keystones in urban spaces. A single mature Peepal tree provides nesting, roosting, and continuous food resources for over 20 species of birds, bats, and insects throughout the dry summer.
- **Specialized Mutualistic Co-Evolution:** The close relationship between the Plain Tiger butterfly and the Rui plant (*Calotropis gigantea*) illustrates co-evolution. The caterpillar's ability to ingest toxic cardiac glycosides protects both the larva and adult butterfly from predatory birds.
- **Anthropogenic Threats to Pollinators:** Honeybees and butterflies face severe mortality from widespread municipal insecticide fogging and decorative lawn pesticide use, which breaks the urban pollination chain.
- **Vulnerability of Exotic Trees:** While Gulmohar is visually stunning, it supported less than one-third of the insect biodiversity found on indigenous Neem and Karanj trees, underscoring why native arboriculture is ecologically superior.

The analysis affirms that conserving urban biodiversity requires deliberate protection of native flora and micro-habitats.

7. Results and Conclusions (निष्कर्ष)

Conclusion: The field study confirms that local neighborhoods harbor rich biological communities capable of thriving alongside urban infrastructure. However, preserving this biodiversity requires conscious protection of indigenous flora, avoidance of chemical insecticides, and safeguarding of wild scrub patches.

Action Checklist for Local Biodiversity Conservation :-

- ❑ Plant only indigenous keystone trees (Neem, Peepal, Banyan, Jamun) on campus and along roadsides.
- ❑ Install shallow clay birdbaths in gardens to provide clean water for avian species during hot summers.
- ❑ Avoid chemical pesticide and herbicide sprays in gardens; utilize natural neem-based pest repellents.
- ❑ Protect wild roadside nectar plants (Calotropis, Tridax, Lantana native) to support butterfly populations.
- ❑ Maintain a personal nature journal to track local seasonal arrivals of migratory birds and insect blooms.

8. Bibliography / References (संदर्भ सूची)

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4. Maharashtra State Board. (2020). *Environmental Education and Water Security (Std. XI)*. Balbharati, Pune.